10.6 Outliers

Lesson Introduction

Benchmark	MA.6.DP.1.6 Given a real-world scenario, determine and describe how changes in data values impact measures of center and variation.	Learning Targets	• I can determine how data is impacted by an outlier. • I can relate the impact of an outlier to which measures of center and variation best describe a distribution.

	Building On	Working Toward
Benchmark Coherence	MA.5.DP.1.2 Interpret numerical data with whole-number values, represented with tables or line plots, by determining the mean, mode, median, or range.	MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.

Essential Questions	• What is an outlier? • How does an outlier affect the median? • How does an outlier affect the mean? • How does an outlier affect the interquartile range? • Are there times when an outlier does not affect the measures?
Lesson Narrative	This lesson begins with students exploring data presented in a line plot. Students analyze the information presented and apply what they know about measures of center and variation to describe and summarize the data, considering the impact an outlier has on the data set and what happens once it is removed. Students summarize the impact of outliers on measures of center and variation, applying this understanding to synthesize data presented in a histogram and another data set.

Component Summary	10.6.1 Warm-Up (~5 min.)	Create meaning from the given story *(SE p. 226)*	10.6.4 Guided Practice (10-15 min.)	Find mean, median, mode, and IQR of data sets *(SE p. 228)*
	10.6.2 Exploration (10-15 min.)	Work with a partner to find the five-number summary and IQR *(SE p. 226)*	10.6.5 Your Turn! (15-20 min.)	Calculate impact of outlier on data set *(SE p. 229)*
	10.6.3 Bring It Together (5 min.)	Synthesize how outliers affect data *(SE p. 228)*	10.6.6 Wrap-Up (~5 min.)	Reflect on understanding of outlier impact on data *(SE p. 230)*

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: • Outlier • Interquartile range	• (Optional) Paper • (Optional) Tape and labels for dot plot • (Optional) Computer	N/A

Teacher Tips

Common Misconceptions	Things to Consider
• Students may incorrectly determine an outlier can *only* be higher or only be lower than the rest of the data. • Students may think that an outlier will always significantly change the mean.	• Consider providing examples and nonexamples of data sets containing outliers to students to decipher the differences. • Allowing students to use calculators during the lesson will help students focus on the effect of outliers on measures of center and variation. Consider have no-calculator questions where students turn their calculator over.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Think-Pair-Share 10.6.4 Guided Practice provides an opportunity for students to engage in a Think-Pair-Share. This routine encourages students to synthesize information as they first consider the question and then discuss with a partner to build onto or revise their thinking. Through this routine, students have the chance to deepen understanding of the impact of outliers.	**MA.K12.MTR.7.1: Real-World Contexts** Students will be making connections between data displayed on graphs and histograms and real-world contexts throughout this lesson. Students will use the information presented to determine measures of center and variation and connect this back to the context of the problems to reason about the impact of outliers when summarizing data distributions.

Differentiate Instruction

Utilizing various display methods to show the data provides a variety of visual entry points for learners. Students can engage with the content through histograms, dot plots, and tables. Encourage students to create their own data displays to help answer questions about the data if you notice them struggling to conceptualize the data presented without a visual display. Students work collaboratively throughout the exploration and guided practice, providing opportunity for mathematical discourse as well as auditory entry points.

Lesson Notes:

10.6.1 Warm-Up: You Tell the Story

Component Overview: *In this Warm-Up, students will engage their critical-thinking skills by reasoning what could have happened in a story. The story will be left open-ended for students to determine what happens.*

10.6.1 Warm-Up: You Tell the Story

1. Robbie goes into a restaurant and orders a deli sandwich and a cola for lunch. Afterward, he pays the bill, tips his waitress, and goes outside. He slowly takes in his surroundings. The sky is black and the city streets are deserted. What happened?
   Answers will vary. The clouds have covered the day light and a storm is coming. People have gone inside because the rain in coming.

Students may get stuck on trying to give a "mathematical" answer. Encourage them to answer using the given information to think critically.

10.6.2 Exploration: Data Values

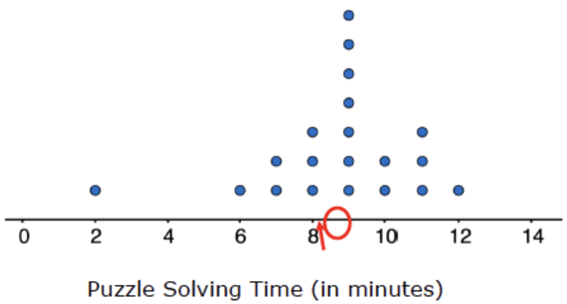
Component Overview: *Students will work with a partner to calculate the five-number summary and IQR of a data set. Students then remove the outliers from the set and recalculate to find the effects an outlier has on the values used to summarize the distribution. Encourage students to make connections between measures that change and do not change when the outlier is removed.*

10.6.2 Exploration: Data Values

Work with your partner to complete the following.

1. Twenty students timed how long it took each of them to solve a puzzle. Below is a list of their solution times, in minutes, and a dot plot displaying the data.

$2, 6, 7, 7, 8, 8, 8, 9, 9, 9, 9, 9, 9, 9, 10, 10, 11, 11, 11, 12$



a. Describe the distribution of data.
Answers will vary. The data is symmetrical if not for the outlier with the value of 2. The outlier causes the data to skew to the left.

b. Determine the mean of the data set. Then, draw an arrow (↑) below the number line that points to this value.
8.7

c. Determine the median of the data set. Then, circle the value on the number line that represents this value.
9

While students are exploring in small groups or pairs, monitor for the use of both informal and formal language in their descriptions. Use their responses as probing questions for 10.6.3 Bring It Together for meaningful synthesis.

For example, you may notice Student A describing the distribution as "normal-looking" or using similar informal language, while Student B in another group may describe it as symmetrical.

As you take note of student responses here, see possible question stems to include in 10.6.3 Bring It Together.

10.6.2 Exploration: Data Values (continued)

d. Discuss with your partner what you notice about the mean and the median of the data. Summarize your discussion.
Answers will vary. The mean and median are very close to each other. The mean is slightly less than the median.

e. What is the range of the data?
10

f. Find the interquartile range (IQR).

Q1	Q3	IQR
8	10	2

2. Imagine that the fastest time of 2 minutes was removed from the data set. Discuss with your partner what effect, if any, removing this value has on the values of center and variation (mean, median, range, IQR). Summarize your discussion.
Answers will vary. Removing the fastest time will make the graph a more accurate description of the time it takes to complete the puzzle.

3. How does the shape of the graph change after 2 is removed?
Answers will vary. The graph would be more symmetrical. The graph will be more representative of the time it takes to complete the puzzle.

4. Compare the new values with the original values.

	Original Value	$>, <, =$	New Value after 2 is Removed
Mean	8.7	$<$	9.1
Median	9	$=$	9
Range	10	$>$	6
IQR	2	$=$	2

Have students make a connection between the outlier and the median and mean.

Notice and Wonder
Implement a Notice and Wonder routine to encourage students to notice what happens to each measure of center and variation after the outliers have been removed.

- *What do you notice about the original values compared to the new value of each category after 2 is removed?*
- *What do you notice about the two categories that stayed the same?*
- *What do you wonder? Do you think this happens when outliers are removed from other sets of data?*

10.6.3 Bring It Together: Outlier Effects

Component Overview: *Students complete statements to describe how the presence of outliers impact the measures of center and variation.*

10.6.3 Bring It Together: Outlier Effects

1. Complete each statement to describe the effect the outlier, 2, has on each measure of center or variation.

   - The presence of an outlier does not have a significant effect on the
     - mean.
     - median.
   - The presence of an outlier will change, often significantly, the value of the
     - mean.
     - median.
   - The presence of an outlier does not have a significant effect on the
     - IQR.
     - range.
   - The presence of an outlier will change, often significantly, the value of the
     - IQR.
     - range.

When outliers are present or a distribution is skewed, the *median* is the preferred measure of center and the *IQR* is the preferred measure of variation to describe the distribution.

Use inquiry and talk moves to have students make sense of their learning with the sentence frames and the blue box, using questions to have them formalize their descriptions from the Exploration.

- *Student A, you described the data distribution as 'normal-looking' – can you say more about what you mean by that?*
- *Student B, how did you describe the distribution?*
- *Student A, based on Student B's response, would you change your answer to the sentence frame? Why or why not?*
- *Student B, what impact does the outlier have on the distribution?*
- *Student A, based on that response, should we use median or IQR in the Exploration? Why?*

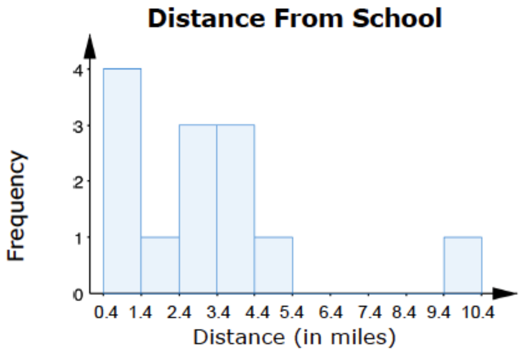
10.6.4 Guided Practice: Data Outliers

Component Overview: *In this Guided Practice, students work with a partner to examine data presented in a histogram. Students answer questions based on the original data set, then examine the differences in the data and measures of center when the outlier is replaced with a data point similar to the rest of the data in the set.*

10.6.4 Guided Practice: Data Outliers

1. Zane and 12 of his friends listed how far, in miles, they live from the school and created a histogram.

$0.4, 0.5, 1, 1, 2.4, 2.7, 3, 3.2, 4, 4.2, 4.4, 5, 9.5$



a. What is the height of the bar for the interval $9.4 - 10.4$?
1

b. Find the mean and the median distance from school.
$3.18, 3$ *respectively*

c. Discuss with your partner what effect that value has on the mean, median, range, and interquartile range of the data. Summarize your conclusions.
Answers may vary. The effect is it skews the data right.

Students may struggle to explain how the outlier impacts each measure of center and variation. Encourage students to use the summary from the Bring It Together as a reference. Further scaffold through questioning, if needed:

- *Do outliers impact the median? What about the IQR? Why?*
- *Do outliers impact the mean? What about the range? Why?*

10.6.4 Guided Practice: Data Outliers (continued)

d. Noelle is Zane's friend who lived 9.5 miles away from the school. She recently moved closer, and now only lives 5.5 miles away from the school. The data set with Noelle's new distance from school, in purple, is shown.

$$0.4, 0.5, 1, 1, 2.4, 2.7, 3, 3.2, 4, 4.2, 4.4, 5, 5.5$$

Would Noelle's new distance from school have similar effects on the data? Explain your reasoning.
Answers may vary. It would shorten the range as 5.5 is closer to the center. The histogram would be more symmetrical.

e. What measure of center best describes the typical distance Zane's friends live from school?
Median

f. What measure of variation best describes the spread of the data distribution?
IQR

Think-Pair-Share
Engage students in a Think-Pair-Share routine before completing Parts d to f. This will provide students with the opportunity to formalize their thinking and synthesize the impact of outliers in a data set and on measures of center and variation.

10.6.5 Your Turn!

Component Overview: *In this component, students will apply their knowledge of calculating the five-number summary and IQR to determine how an outlier can impact the data. Students remove an outlier to determine how it impacts a data set and the measures of center and variation.*

10.6.5 Your Turn!

1. Zoe needed exact change to purchase a snack and a drink from the vending machine. She asked her friends how much change, in dollars, they had and listed the results.

   $0.15, 0.15, 0.20, 0.25, 0.25, 0.30, 0.30, 0.30, 0.35, 0.75, 0.85$

   a. What do you notice about the data?
   Answers will vary. All numbers are less than $\$1$.

   b. Will the mean be higher, lower, or approximately the same as the median? Explain your reasoning or justify your answer by showing your work.
   Answers may vary. Slightly higher. Mean is 0.35 and the median is 0.30.

   c. Describe the impact of removing 0.85 from the data set.
   Answers may vary. The data becomes more symmetrical. Now 0.75 is the only outlier. The mean will decrease.

   d. The values 0.85 and 0.75 are outliers in this data set. What happens to the range if 0.85 and 0.75 were both removed from the data set?
   The range becomes smaller and the spread of the data is smaller

   e. The IQR of the full data set is 0.15.

   The IQR with 0.85 and 0.75 removed is 0.125.

   f. Is there a big difference between the IQR of the original data set and the modified data set?
   No, the IQR is not significantly affected by the removal of outliers.

Consider providing a graphic organizer or table for students to use when finding the mean, median, and interquartile range.

To incorporate digital learning, allow students to type the numbers into a column on an Excel spreadsheet to manipulate the data and view how changing the different results changes the data.

10.6.6 Wrap-Up: Cool Down

Component Overview: *This Wrap-Up assesses students' abilities to determine if and how an outlier affects different measures of center and variation for a set of data.*

10.6.6 Wrap-Up: Cool Down

1. The presence of an outlier has a significant effect on which measures of center and spread? Select all that apply.

   - ☐ Mean
   - ☐ Median
   - ☐ Interquartile Range
   - ☐ Range

2. A data set has an outlier that is significantly less than most of the other values.

   a. Explain how this impacts the measures selected in number 1.
   A lower outlier will cause the mean to be smaller than the median and will increase the range of the data.

   b. Which measures of center and variation would you use to best describe the data?
   Answers may vary. Median and IQR (interquartile range).

Consider using all or parts of this exercise as a formative assessment to gather data on student progress from this lesson and inform groupings for remediation or partner work in the next lesson.